

FGV EAESP — SÃO PAULO SCHOOL OF BUSINESS ADMINISTRATION

MSC / PHD IN BUSINESS ADMINISTRATION

Time Series Econometrics for Business Research

Lecture 2

Non-Stationarity: Unit Roots and Trends

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Topics covered: Integrated Processes · Random Walks · Spurious Regression ·
Trend-Stationarity vs. Difference-Stationarity · Dickey–Fuller Tests · ADF · Phillips–Perron
· KPSS · Structural Breaks and Unit Roots

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1. Motivation: Why Stationarity Matters

The ARMA framework developed in Lecture 1 rests on the assumption of covariance stationarity. Most macroeconomic and financial series — price levels, exchange rates, GDP, consumption, and many marketing metrics measured over long horizons — do not satisfy this assumption. Applying ARMA methods directly to non-stationary series produces unreliable estimates, invalid t -statistics, and meaningless long-run forecasts.

- (i) OLS t -statistics do *not* follow t -distributions; inference is invalid.
- (ii) Regressions between independent non-stationary series produce spuriously significant results.
- (iii) Forecast uncertainty grows without bound as the horizon increases.
- (iv) Standard ACF/PACF identification is distorted: the ACF decays very slowly.

2. Integrated Processes and the Random Walk

2.1. The random walk

Definition 2.1. A series Y_t follows a *random walk* if

$$Y_t = Y_{t-1} + \varepsilon_t, \quad \varepsilon_t \stackrel{iid}{\sim} (0, \sigma^2), \quad Y_0 = 0.$$

Since $Y_t = \sum_{s=1}^t \varepsilon_s$, the properties are:

$$\begin{aligned} E(Y_t) &= 0, \\ \text{Var}(Y_t) &= t\sigma^2 \quad (\text{grows without bound}), \\ \text{Cov}(Y_t, Y_{t+s}) &= t\sigma^2 \quad (\text{depends on } t, \text{ not just } s). \end{aligned}$$

The variance grows linearly with t — a fundamental violation of covariance stationarity. Wold's decomposition fails because the $\text{MA}(\infty)$ coefficients are all unity: past shocks have *permanent* effects on the level of Y_t .

A random walk has $\hat{\rho}(1) \approx 0.99$, as does a stationary AR(1) with $\phi = 0.99$. The two processes behave fundamentally differently for inference and forecasting. Formal unit root tests are required.

2.2. Random walk with drift

$$Y_t = \mu + Y_{t-1} + \varepsilon_t \implies Y_t = \mu t + \sum_{s=1}^t \varepsilon_s.$$

Adds a deterministic linear trend on top of stochastic wandering. Typical for nominal asset prices, log GDP, and log CPI. Mean $E(Y_t) = \mu t$ grows with t ; the series is still non-stationary.

2.3. Integrated processes: $I(d)$

Definition 2.2. A series Y_t is **integrated of order d** , written $Y_t \sim I(d)$, if it requires differencing d times to achieve covariance stationarity:

$$\Delta^d Y_t = (1 - L)^d Y_t \sim I(0).$$

Order	Treatment	Typical examples
$I(0)$	None	Returns, spreads, stationary residuals
$I(1)$	First difference	Log prices, exchange rates, log CPI, log GDP
$I(2)$	Second difference	Log nominal wages in some episodes

Unit root and AR representation. An AR(1) $Y_t = \phi Y_{t-1} + \varepsilon_t$ has characteristic equation $1 - \phi z = 0$.

- $|\phi| < 1$: root outside unit circle $\Rightarrow$ stationary $I(0)$.
- $\phi = 1$: root *on* unit circle $\Rightarrow I(1)$ — the unit root case.
- $|\phi| > 1$: explosive.

For an AR(p) with a unit root: $\phi(L) = (1 - L)\tilde{\phi}(L)$ where all roots of $\tilde{\phi}(L) = 0$ lie outside the unit circle.

3. Spurious Regression

Generate two *independent* random walks Y_t and X_t . Regress Y_t on X_t :

$$Y_t = \alpha + \beta X_t + u_t.$$

Despite $\text{Cov}(\varepsilon_t^Y, \varepsilon_t^X) = 0$, the t -statistic on $\hat{\beta}$ exceeds 1.96 in approximately **75% of Monte Carlo replications**. R^2 is high; the Durbin–Watson statistic is near zero.

Why spurious regressions occur. Both series share a common stochastic trend. OLS mistakes this shared non-stationarity for a structural relationship. Under $H_0 : \beta = 0$ with $I(1)$ variables, t -statistics *diverge* as $T \rightarrow \infty$ rather than converging to a t -distribution.

In a spurious regression:

- Residuals are non-stationary (ADF on $\hat{u}_t$ fails to reject unit root).
- Durbin–Watson ≈ 0 .
- $R^2 \rightarrow 1$ as $T \rightarrow \infty$ even though the true relationship is absent.

Whenever $R^2 > DW$, suspect spurious regression.

4. Trend-Stationarity vs. Difference-Stationarity

Two series may both exhibit trending behaviour yet require fundamentally different treatments.

Trend-Stationary (TS)	Difference-Stationary (DS)
$Y_t = \mu + \delta t + u_t, u_t \sim I(0)$	$Y_t = \mu + Y_{t-1} + \varepsilon_t$
Deterministic trend	Stochastic trend
Shocks are <i>transitory</i> : deviations from trend are temporary	Shocks are <i>permanent</i> : level never reverts
Correct treatment: detrend	Correct treatment: difference
After detrending: $\hat{u}_t \sim I(0)$	After differencing: $\Delta Y_t = \mu + \varepsilon_t \sim I(0)$

Detrending a DS series (applying OLS on t) leaves non-stationary residuals — spurious detrending. **Over-differencing a TS series** introduces a non-invertible MA unit root, distorting all subsequent inference.

Nelson and Plosser (1982). Testing 14 US macroeconomic series over 1909–1970, they could not reject the unit root null for 13 out of 14 series (exception: unemployment rate). They concluded that most macro series are DS, implying that real and monetary shocks have *permanent* effects on output.

5. The Dickey–Fuller Family of Tests

5.1. The Dickey–Fuller test

Definition 5.1. Subtract Y_{t-1} from both sides of $Y_t = \phi Y_{t-1} + \varepsilon_t$:

$$\Delta Y_t = \pi Y_{t-1} + \varepsilon_t, \quad \pi = \phi - 1.$$

$H_0 : \pi = 0$ (unit root) vs. $H_1 : \pi < 0$ (stationary).

Three standard specifications:

- (i) No constant, no trend: $\Delta Y_t = \pi Y_{t-1} + \varepsilon_t$.
- (ii) Constant, no trend: $\Delta Y_t = \alpha + \pi Y_{t-1} + \varepsilon_t$.
- (iii) Constant and trend: $\Delta Y_t = \alpha + \delta t + \pi Y_{t-1} + \varepsilon_t$.

Under H_0 , the t -statistic on $\hat{\pi}$ does **not** follow a t -distribution. It follows a Dickey–Fuller distribution:

$$\tau \xrightarrow{d} \frac{\int_0^1 W dW}{\left(\int_0^1 W^2\right)^{1/2}},$$

where W is a standard Brownian motion. This is left-skewed with critical values more negative than -1.96 . At 5%: -1.95 (no constant), -2.86 (constant), -3.41 (constant + trend).

5.2. Augmented Dickey–Fuller (ADF) test

Add lagged differences to control for serial correlation in residuals:

$$\Delta Y_t = \alpha + \delta t + \pi Y_{t-1} + \sum_{j=1}^p \gamma_j \Delta Y_{t-j} + \varepsilon_t.$$

Test statistic: the t -ratio on $\hat{\pi}$. The same DF critical values apply. Lag length p is selected by BIC or sequential t -testing starting from $p_{\max}$.

- (i) Plot the series. If it wanders around a fixed level $\Rightarrow$ include a constant.
- (ii) If it shows a clear trend $\Rightarrow$ include constant and trend.
- (iii) Over-specifying deterministics reduces power but avoids size distortion. Under-specifying (omitting a constant when there is drift) severely inflates size.

5.3. Phillips–Perron (PP) test

Rather than adding lagged differences (as in ADF), PP applies a non-parametric correction to the t -statistic using a Newey–West long-run variance estimator:

$$\hat{\omega}^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^q \left(1 - \frac{j}{q+1}\right) \hat{\gamma}_j.$$

Advantages: robust to heteroskedasticity; no lag length selection. Disadvantage: severe size distortions when errors are highly autocorrelated. ADF is generally preferred in finite samples.

5.4. KPSS test

Kwiatkowski, Phillips, Schmidt & Shin (1992) *reverse* the null hypothesis:

$$H_0 : Y_t \sim I(0) \quad \text{vs.} \quad H_1 : Y_t \sim I(1).$$

Definition 5.2. Decompose $Y_t = \xi t + r_t + \varepsilon_t$ where $r_t = r_{t-1} + u_t$. Test $H_0 : \sigma_u^2 = 0$. The KPSS statistic is:

$$\eta = \frac{1}{T^2 \hat{\omega}^2} \sum_{t=1}^T S_t^2, \quad S_t = \sum_{s=1}^t \hat{e}_s,$$

where $\hat{e}_t$ are OLS residuals from regressing Y_t on deterministics.

ADF result	KPSS result	Conclusion
Reject H_0	Do not reject H_0	Strong evidence for $I(0)$
Do not reject H_0	Reject H_0	Strong evidence for $I(1)$
Both reject	—	Conflicting evidence
Neither rejects	—	Insufficient power

6. Structural Breaks and Unit Root Tests

6.1. Perron (1989): the interaction between breaks and unit roots

Perron showed that *unmodelled structural breaks* reduce the power of ADF tests. A stationary series with a level shift will look like a random walk to the ADF test because the break mimics the behaviour of an integrated process.

Re-testing the Nelson–Plosser 14 series with dummy variables for the Great Crash (1929) and the 1973 oil shock, Perron (1989) found that **11 out of 14** series rejected the unit root null — the opposite conclusion from Nelson–Plosser. The treatment of structural breaks matters enormously.

6.2. Zivot–Andrews (1992) test

Allows for an endogenous break date by computing the ADF statistic for every possible break point and taking the minimum:

$$\inf_{t^*/T \in (0,1)} \tau(t^*).$$

Critical values are more negative than standard ADF values to account for the data-mining over break dates.

7. Practical Guidelines for Unit Root Testing

1. **Plot the series.** Identify trend, level shifts, or structural changes visually.
2. **Choose specification.** Based on the plot, select constant/trend. Do not data-mine the specification.
3. **Select lag length.** Use BIC or sequential t -testing on γ_j in ADF.
4. **Run ADF and KPSS.** Cross-check conclusions.
5. **Test for structural breaks.** If suspected, use Zivot–Andrews or Perron (1989).
6. **Report all specifications.** Transparency is essential.

Over-differencing wastes information (at most one observation lost) and introduces an MA unit root that slightly inflates standard errors. Under-differencing (treating $I(1)$ as $I(0)$) produces invalid inference throughout the entire model. The costs are asymmetric: when in doubt, difference.

8. Applied Examples

8.1. S&P 500 log price

The log S&P 500 index is a canonical $I(1)$ series. ADF (constant, no trend, BIC-selected lags) fails to reject the unit root null. KPSS (level stationary null) rejects strongly. First-differencing yields log-returns that are approximately $I(0)$, with ACF/PACF structure consistent with weak serial correlation — motivating the ARMA analysis of Lecture 1.

8.2. USD/BRL exchange rate (continued from Lecture 1)

Continuing the exchange rate example: ADF on $\log(\text{Dolar})$ fails to reject a unit root (test statistic ≈ -1.8 , 5% CV = -2.86). KPSS rejects the stationarity null. After first-differencing, $\Delta \log(\text{Dolar})$ is stationary, confirming the $I(1)$ conclusion adopted in Lecture 1.

8.3. Business application: long-run advertising–sales relationship

If monthly sales S_t and advertising spend A_t are both $I(1)$, a regression $S_t = \alpha + \beta A_t + u_t$ estimated by OLS may be spurious unless the two series are cointegrated. The next step — testing for cointegration and estimating an Error Correction Model — is the subject of Lecture 3.

9. Core References

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